THE DEVELOPMENT OF PENETRATION TESTING SYSTEM FOR CYBERSECURITY ATTACK

NATTAPONG SEEMOONPA
THONGTHAE SUWANMANASILN
HEMKITTI NORKAEWCHAROEN

BACHELOR OF ENGINEERING

IN

COMPUTER ENGINEERING

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A SENIOR PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
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EXAMINING COMMITTEE

....ADVISOR
(Dr.Mahamah Sebakor)

....COMMITTEE
(Asst. Prof. Gp.Capt. Dr.Thongchai Yooyativong)

....COMMITTEE

(Asst. Prof. Dr.Roungsan Chaisricharoen)

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NATTAPONG SEEMOONPA
THONGTHAE SUWANMANASILN
HEMKITTI NORKAEWCHAROEN

Title	THE DEVELOPMENT OF PENETRATION TESTING SYSTEM FOR CYBERSECURITY ATTACK
Author	Mr. Nattapong Seemoonpa Mr.Thongthae Suwanmanasiln Mr.Hemkitti Norkaewcharoen
Degree	Bachelor of Engineering (Computer Engineering)
Supervisory Committee	
Advisor	Dr.Mahamah Sebakor
Committee	Asst. Prof. Gp.Capt. Dr.Thongchai Yooyativong
Committee	Asst. Prof. Dr.Roungsan Chaisricharoen

Abstract

For three years we have studied in the School of Information Technology, Computer Engineering major. It Makes us interested in Network security
Nowadays, there are more cyber threats, so we will create a sandbox area to develop cybersecurity skills for prepare to handle and prevent the cyber threats in the future that may occur to ourselves or our company and prevent the other people or bad intentions who want to hack the company or reduce the damage that may occur in a wide range to be less damage.In addition, we're using the sandbox to practice threat defense, and we will use Docker to help simulate any problems. This makes data protection more solid and secure.The Hacker side is for detecting abnormal things. Look at your own or company's mistakes and fix them on the spot.

Keyword: The development of penetration testing system for cybersecurity attack

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CHAPTER 1
INTRODUCTION

1.1 Background and Rationale

Recently, Sandbox has been used in Information Technology, especially Network Security.It is used to test network systems in a limited scope. There is also Docker to help simulate events when running a Service or Server to easily handle problems that will happen.In the network system have many users. It has both good intentions or threats and the most common threat is a Hacker, a person who has computer expertise in hacking a computer system through a network, thus creating a job called Network Security.It is another profession that is important in the field of Network to protect systems and important information for customers or companies. So our project will train and simulate our hacking by using SandBox as an experiment via VMware.

1.2 Objective

1.2.1 To see the mistakes of yourself or the company and fix the problem immediately.

1.2.2 learning relate web application for security
1.2.3 Try to hacking lab from easy to supreme
1.2.4 use many tools for hacking advantage

1.3 Scope

1.3.1 MFU user

1) This system can be used at Mae Fah Luang University only.
2) Users must be students or Mae Fah Luang personnel only.

1.3.2 Administrator

1) List all administrator's

1.3.3 Create problems 30+
1.3.4 Use many tools for hacking such as sqlmap, nmap, jhon the ripper,burp suite, invicti, fortify weblnspect, cain&abel, nessus, nikto, acunetix, intruder, metasploit etc.
1.3.5 Techniques such as basic command linux, brute force, privilege escalation, reverse shell, sql injection , xss(cross site scripting),local file inclusion, etc.

1.4 Methodology

1.4.1 Proposal preparation and defense

1.4.2 Install VMware,Kali Linux and Docker

1.4.3 Research and study about the required programs and network systems.

1.4.4 Simulate the problem situation and try to practice hacking into the network system.

1.4.5 Senior project 1 document preparation

1.4.6 Senior Project 1 exam

1.4.7 System development

1.4.8 System testing

1.4.9 Senior Project 2 exam

1.4.10 Senior Project 2 exam

1.5 Plan

Table 1.1 Working plan

	Senior project 1					Senior project 2				
Plan/Month	Jan	Feb	Mar	Apr	May	Aug	Sep	Oct	Nov	Dec
Proposal preparation and defense										
Literature review and requirement gathering										
System analysis										
System design										
Senior project 1 document preparation										
Senior Project 1 exam										
System development										
System testing										
Senior project 2 document preparation										

Senior Project 2 exam										

1.6 Expected Result

1.6.1 Develop network security skills and understand the working system of the network.

1.6.2 Can apply the knowledge to develop the skills and use in future work

1.7 Place and Equipment

1.7.1 Place

- In MFU

1.7.2 Hardware

- Laptop
- Computer

1.7.3 Software

- Vmware Workstation
- Ubuntu
- PuTTY

CHAPTER 2
LITERATURE REVIEW

2.1 Related Theory

Cybersecurity is a multidimensional field with various theories and models aimed at protecting data and information systems from cyber threats. Fundamental principles, such as confidentiality, integrity, and availability, form the basis for organizational policies. Survival models, maturity models, and defense-in-depth strategies contribute to predicting, planning, and implementing effective responses to cyber threats. Addressing insider threats, establishing strong security governance, considering geopolitical factors, and advancing cybersecurity technologies are integral aspects.

Overall, these theories guide the formulation of policies and measures to prevent and mitigate potential cyber threats for organizations.

2.2 Related Research

Penetration Testing, also known as Pentest, is a process in which cyber security professionals attempt to penetrate an organization's systems or networks to identify vulnerabilities that can be exploited.

Penetration testing aims to help organizations understand their security risks and improve their defenses. By identifying vulnerabilities that can be exploited Organizations can take steps to patch those vulnerabilities and reduce their risk of attack.

Penetration testing is often performed by external cybersecurity experts. However, some organizations may choose to conduct internal penetration testing using their own cybersecurity personnel.

Here are some examples of related research on penetration testing in Thai.

Research on "Application of Penetration Testing to Assess Computer Network Security" by Dr. Somchai Muangchan and colleagues from Rangsit University. Study the application of penetration testing to assess the security of computer networks. By comparing test results from various methods such as black box testing. White box testing and gray box testing.

2.3 Related Work

Pentest or Penetration Test is a system penetration test. Or what is colloquially known as Ethical Hacking It is a simulated cyberattack that is authorized on a computer system. This is done to assess the security of the system. This should not be confused with vulnerability assessment. Testing is carried out to identify weak points.

2.4 Related Technology

2.4.1 VMware Workstation

Vmware Workstation is a personal computer virtualization software that allows users to create and run virtual operating systems on the same computer.

Figure 2.4.1 Logo VMware Workstation image

2.4.2 Ubuntu is a free and open-source operating system based on the Linux kernel. It is a community-developed project made available under the terms of the GNU General Public License.

Figure 2.4.2 Logo Ubuntu image

2.4.1 PuTTY is a free and open-source terminal emulator, serial console, and network file transfer application. It supports a wide range of network protocols, including SCP, SSH, Telnet, rlogin, and raw socket connections. It can also connect to serial ports.

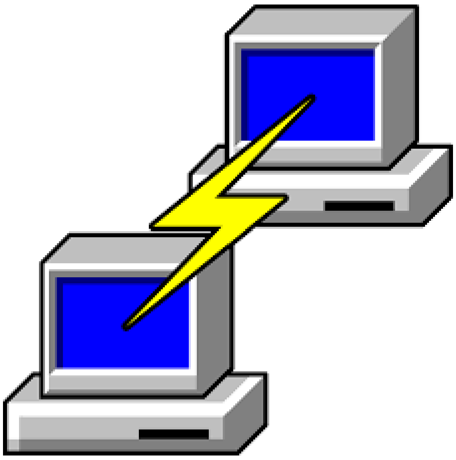
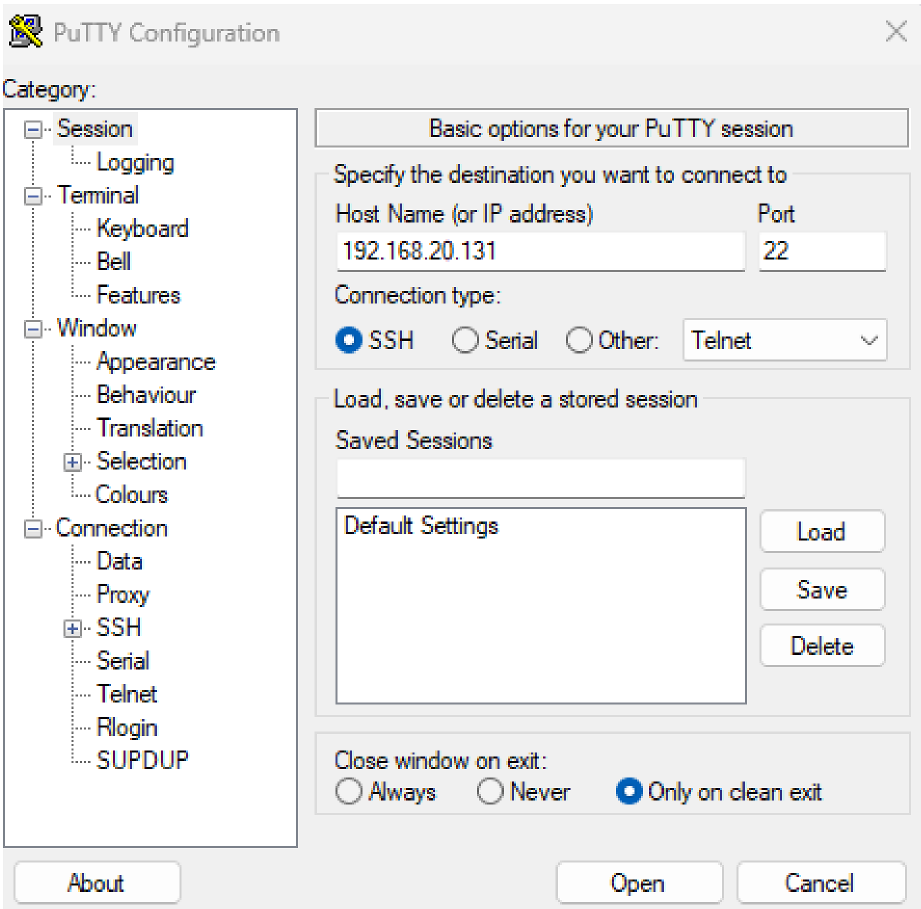


Figure 2.4.3 Logo PuTTY image

CHAPTER 3
METHODOLOGY

3.1 Connect SSH

3.1.1 Connect PuTTY, add IP 192.1688.20.131 and Port 22, Connection Type: SSH



3.1.1 PuTTY image

3.1.2 That will show the terminal, open The Problem, First to check the current directory “pwd” , By “cat Problem” that will show that problem.

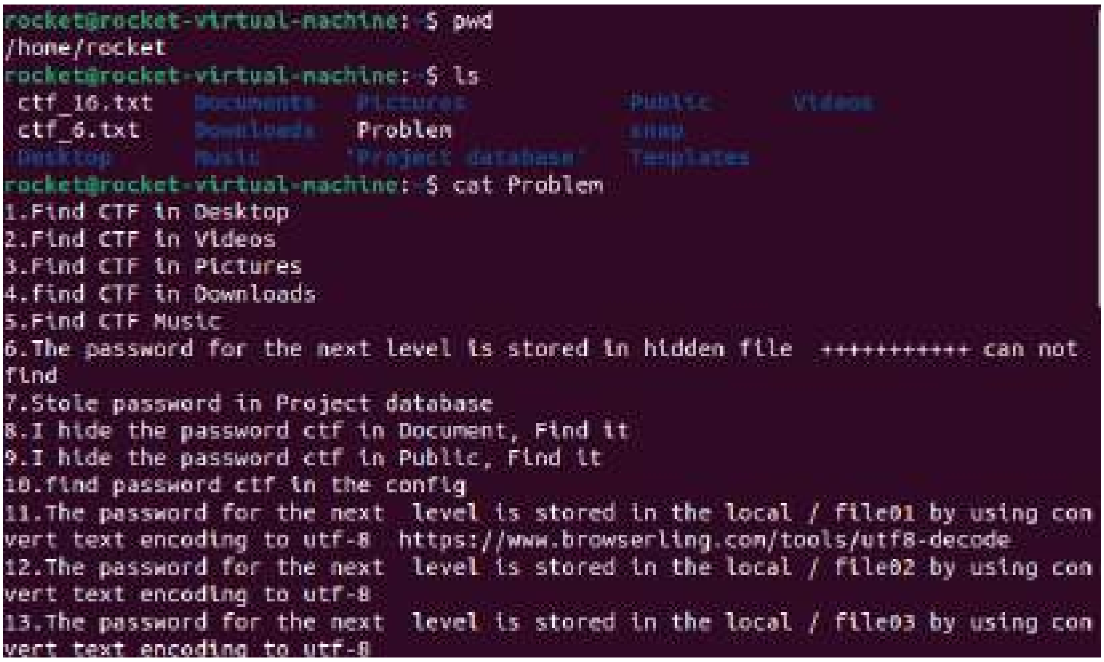


Figure 3.1.2 Problem image

3.2 Practice solving problems

Practice problem solving

Practice solving problems with different levels of difficulty according to the questions.

The player must study and learn how to use it to issue commands to solve problems. When faced with various problems

After players understand the commands, log in to the terminal to begin testing their knowledge and understanding of the commands.

3.3 Solve problems by yourself

Solve the problem yourself

Self-troubleshooting: We hide problems in files and folders. You will need to find a way to get into the folder where the flag files are hidden. You can find knowledge in solving problems from various sources such as picoCTF, DropCTF, bandit, etc.

Figure 3.3.1 picoCTF image

Figure 3.3.2 DROPCTF image

Figure 3.3.3 Over The Wire BANDIT image

CHAPTER 4

EXPERIMENT AND RESULTS

In the capture the flag project, cybersecurity knowledge is used to solve problems to find hidden answers.

4.1 Basic command

We hide and seek flag in Desktop, you should to check flag
First “cd” for enter folder
and “ls” for list everything file or folder in this directory
Finally, if you detect a flag, use “cat” to show and read file text.

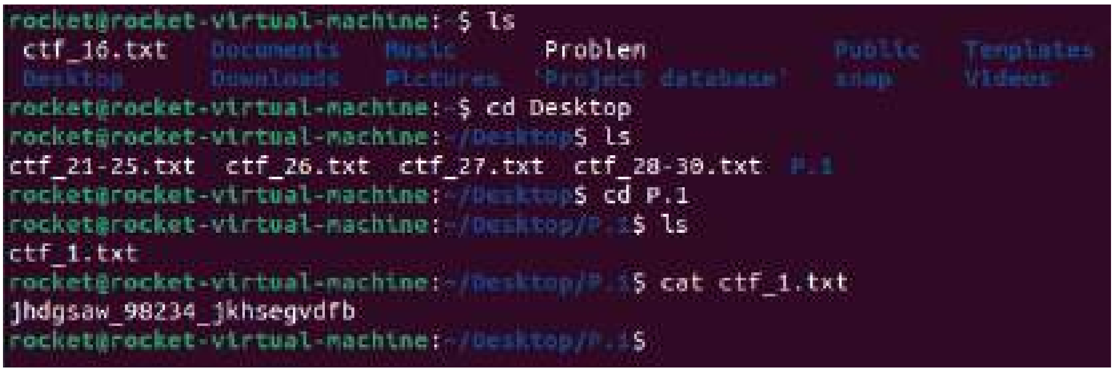


Figure 4.1.1 Basic command image

4.2 Intermediate

Use command “ls -alps” that will show file or folder or directory because “mozilla/ Iron9” normally can not see that hide and seek.

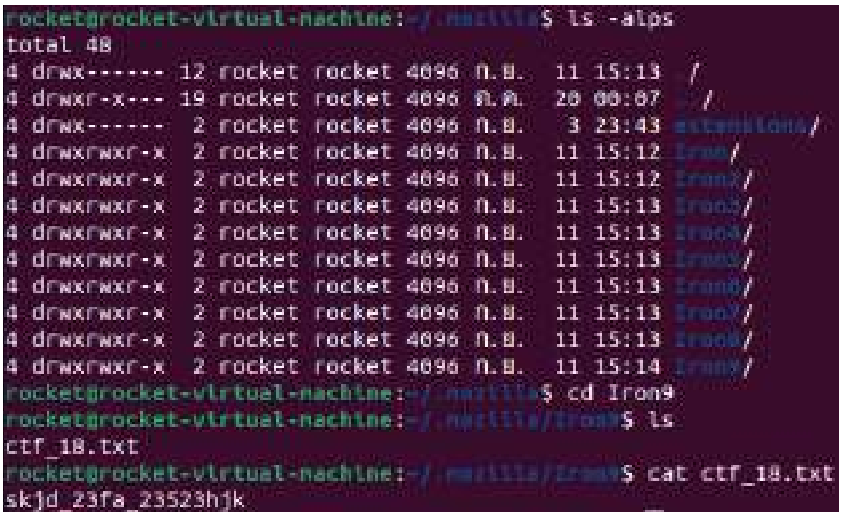


Figure 4.2.1 Intermediate image

4.3 Advanced

4.3.1 If you want to find some word but have many word and have no time to much you can use “strings “name file” | grep “word you want to find” ”

Figure 4.3.1 Advanced image

4.3.2 If you want to find some word but there are a lot of duplicates. and have no time to much you can use “sort “name file” | uniq”

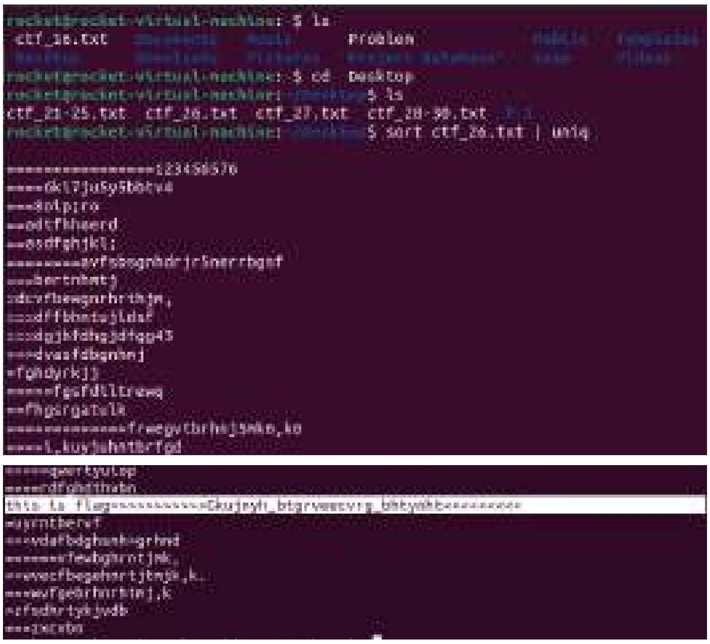


Figure 4.3.2 Advanced image

CHAPTER 5
CONCLUSION

Conclusion

Penetration testing is an essential activity in ensuring the security of computer systems, networks, and applications. It is a process of identifying and exploiting vulnerabilities in a system to determine its level of security and to provide recommendations for improving it. As cyber threats become more sophisticated, it is crucial to develop robust and effective penetration testing systems to protect against cyber-attacks.In recent years, there has been a significant increase in the development of penetration testing systems for cybersecurity attacks. These systems have become more advanced and complex, incorporating a wide range of techniques and tools to identify and exploit vulnerabilities in computer systems. One of the main challenges in developing a penetration testing system is to ensure that it is effective and reliable. This requires a thorough understanding of the different techniques used by cybercriminals and the vulnerabilities they target. Additionally, it is essential to keep the penetration testing system up-to-date with the latest security patches and updates. Another critical aspect of developing a penetration testing system is to ensure that it is user-friendly and accessible. The system should be easy to use, even for non-technical users, and provide clear and concise reports on the vulnerabilities identified and recommendations for remediation.

In conclusion, the development of penetration testing systems is crucial for ensuring the security of computer systems and networks against cyber-attacks. As cyber threats continue to evolve, it is essential to develop robust and effective penetration testing systems that can identify and exploit vulnerabilities in a system and provide recommendations for improving its security.

5.1 Discussion

Penetration testing is a critical component of cybersecurity as it helps identify and mitigate potential vulnerabilities in computer systems, networks, and applications. By simulating attacks on a system, organizations can identify and address potential weaknesses before they are exploited by malicious actors.The development of effective penetration testing systems is an ongoing challenge for cybersecurity professionals. Cybercriminals are continually developing new and sophisticated techniques to bypass security measures, and so, penetration testing systems need to keep pace with these evolving threats.One of the key challenges in developing a penetration testing system is ensuring that it is accurate and reliable. To achieve this, the system must use the latest tools and techniques, be well-designed, and effectively simulate real-world scenarios. Additionally, it is crucial to test the system thoroughly to ensure that it is detecting all potential vulnerabilities and that the results are accurate.Another critical consideration is the usability of the penetration testing system. The system should be easy to use, with clear and concise reports that are easily understandable, even by non-technical users. This is particularly important when organizations need to communicate the results of penetration testing to stakeholders or senior management.Finally, as the use of cloud services and mobile devices continues to increase, penetration testing systems need to adapt to these changing environments. This requires an understanding of how cloud and mobile technologies are used and how they can be exploited by cybercriminals.In conclusion, the development of a robust and effective penetration testing system is critical for ensuring the security of computer systems and networks. This requires a thorough understanding of the latest tools and techniques used by cybercriminals, as well as an awareness of the changing landscape of cloud and mobile technologies. Ultimately, a well-designed and accurate penetration testing system can help organizations proactively identify and mitigate potential vulnerabilities before they are exploited by malicious actors.

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